STORMIT · HANDOVER PRESENTATION

Serverless Knowledge Assistant

Engineers ask the AWS Serverless Applications Lens questions in plain English and get answers grounded in the whitepaper, with page citations. Built on Amazon Bedrock and S3 Vectors, deployed with Terraform.

October 2026 · Sereda Artur

Functional Architecture

INGEST · RUNS WHEN A PDF IS UPLOADED

1 · Upload
PDF lands in the private docs bucket

→

2 · Extract
pypdf reads the text page by page

→

3 · Chunk
1,000 chars, 200 overlap

→

4 · Embed
Titan Text Embeddings V2, 1,024 dims

→

5 · Store
Vector, text and metadata in S3 Vectors

ASK · RUNS PER QUESTION

1 · Ask
Question from the web UI

→

2 · Embed
Same Titan model as ingest

→

3 · Retrieve
Top 5 chunks by cosine distance

→

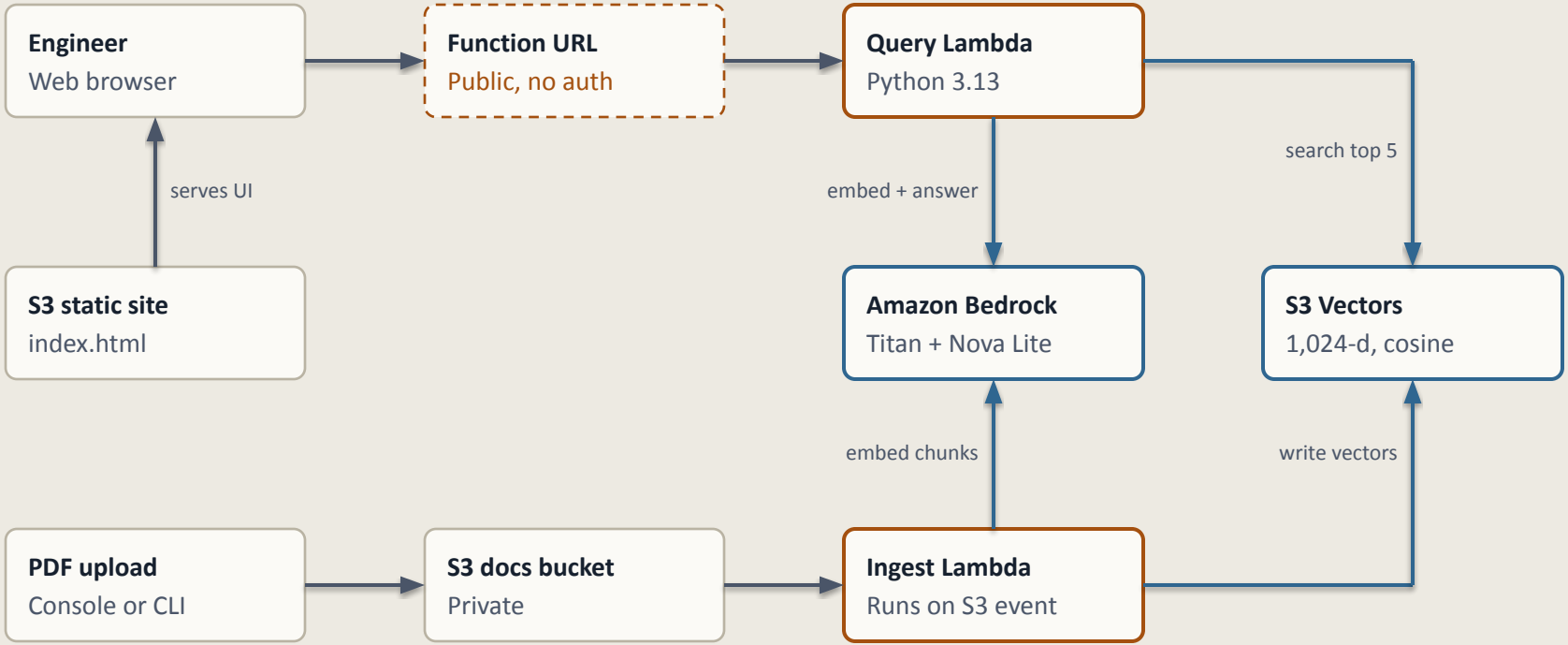
4 · Ground
Prompt: answer only from context, cite [n]

→

5 · Answer
Nova Lite at temperature 0, with sources

Out of scope by design (SoW): authentication, CI/CD, framework frontend, managed cluster

Cloud Architecture on AWS



Terraform-managed · IAM role per function, scoped to named ARNs · CloudWatch Logs (7 days) · us-east-1 by default

PoC Findings

207
chunks from the 104-page whitepaper, measured with the PoC chunker

1
command to deploy: terraform apply builds, provisions and seeds the PDF

0
always-on resources: everything bills per request or per GB stored

- **S3 Vectors suits a PoC:** no cluster to run and no minimum spend, so idle cost is close to storage only.
- **Answers stay grounded:** a context-only prompt at temperature 0, with page citations the reader can check.
- **Extraction limits quality:** chunks fill to the size cap (average 981 chars), so they ignore section breaks, and tables lose their structure.

Gap Analysis

Area	Status	Gap or caveat
Deployment	Caveat	Packaging calls python; hosts with only python3 need an alias
Functionality	Met	PDF text only: tables and diagrams are flattened or lost
Security (IAM)	Met	No admin rights; but the API is public and the site is HTTP only
Documentation	Met	README covers deploy, outputs, variables and tear-down
Data lifecycle	Open	Deleting or shrinking a PDF leaves stale vectors in the index
Scale	Open	One embedding call per chunk; the 300 s timeout caps PDF size
Operations	Open	Local Terraform state, no alarms or DLQ, raw errors sent to caller
Answer quality	Open	No evaluation set to measure accuracy or catch regressions

The first four rows are the SoW success criteria. The rest are gaps to close before a pilot.

Recommendations: PoC to Pilot

1 · HARDEN

Before wider use

- CloudFront with HTTPS in front of the site and API
- Authenticate the API with IAM auth or Cognito
- Delete vectors when a PDF is removed or replaced
- Generic error responses, detail kept in logs
- Remote Terraform state with locking

2 · IMPROVE

Better answers

- A golden question set, scored on every change
- Chunk by section headings, not character count
- Table and figure extraction, e.g. Amazon Textract
- Compare Nova Lite with a larger Bedrock model

3 · SCALE

Operate at volume

- Parallel ingestion via SQS, with a DLQ
- CloudWatch alarms and an AWS Budgets limit
- CI/CD for Terraform and Lambda code build
- An EU region for data residency
- Assess Bedrock Knowledge Bases as the managed path